

SOFTWARE ENGINEERING LAB – 2

(UE24CS341A)

Agile Backlog Creation & Sprint Simulation in Jira

Team Number: 001

Individual Submission: Mohammed Arfan Asgar (PES1UG24AM163)

Problem Statement 1: Smart Lab Equipment & Slot Reservation Portal

EPIC: Epic 1 – Equipment Slot Reservation (Mapped Requirements FR-001, FR-002, FR-003)

Note: This report covers only Epic 1, my individual contribution from the Lab 1 report. The remaining requirements are covered separately by my teammates.

Lab Objective

This lab converts the requirements FR-002 and NFR-002 from the Lab 1 report into an Agile Epic and User Stories in Jira, estimates effort using the Fibonacci scale, prioritises the backlog, simulates two one-week sprints on a Scrum board, and tracks sprint progress using the Burndown Chart.

1. Epic (Derived from FR-002 and NFR-002)

Epic Key	Epic Name	Mapped Requirements	Description
SEL2-1	Epic 1: Reserve Equipment Slot	FR-001, FR-002, FR-003, NFR-001, NFR-002	Enable authenticated students to view available laboratory equipment and reserve valid equipment slots while checking availability, calibration status, reservation limits, and preventing overlapping reservations. The system should process concurrent reservation requests securely and preserve confirmed reservation records and tokens.

2. User Stories for Epic 1

The Epic is broken into 6 User Stories

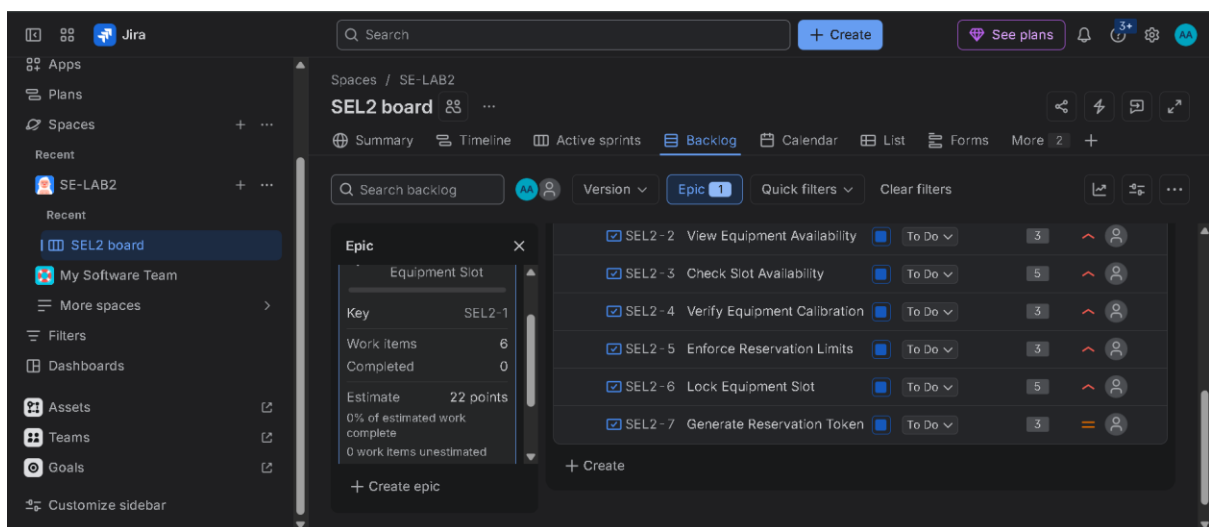
Story User Story		Points
1	View Equipment Availability — As a student, I want to view available laboratory equipment and time slots so that I can choose a suitable equipment slot for reservation.	3
2	Check Slot Availability — As a student, I want the system to check whether my selected slot is available so that I do not make a conflicting reservation.	5
3	Verify Equipment Calibration — As a student, I want the system to verify equipment calibration before reservation so that I can reserve only valid equipment.	3
4	Enforce Reservation Limits — As a student, I want the system to enforce the allowed reservation duration and advance-booking limits so that I follow laboratory booking rules.	3
5	Lock Equipment Slot — As a student, I want the selected equipment slot to be locked during reservation so that another student cannot reserve the same slot simultaneously.	5
6	Generate Reservation Token — As a student, I want a unique reservation token after successful booking so that I have confirmation of my reservation.	3

Total Backlog estimate = 22 Story Points

3. Jira Backlog – Epic, User Stories , Story Points and Backlog Prioritization

The screenshot shows Epic 1 with all six User Stories, the priority icon and story points on each row, and the Epic panel reporting 6 work items and an estimate of 22 points.

And also shows Backlog Prioritization User Stories.



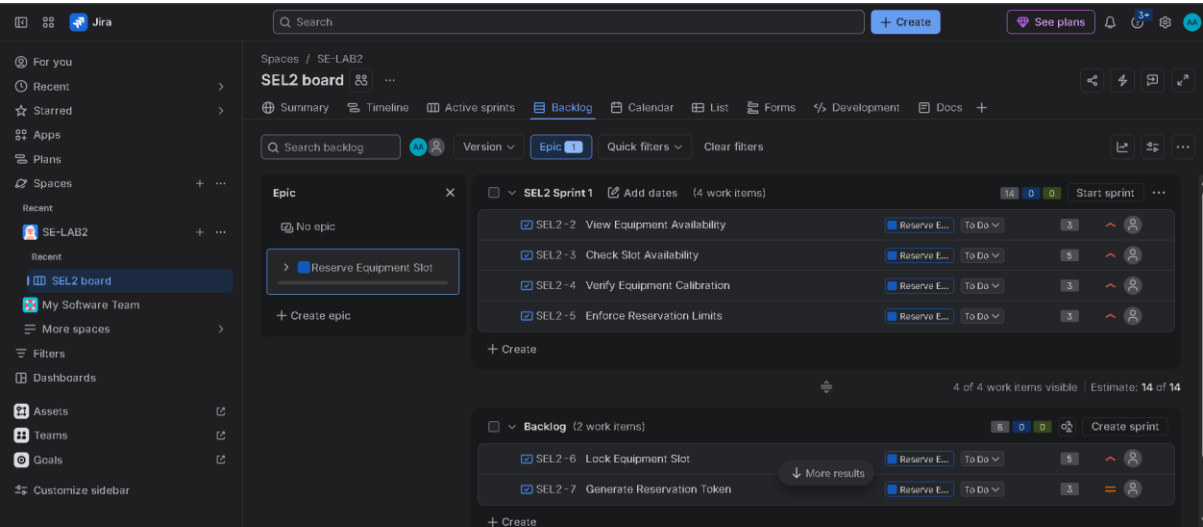
Total Story Points (Epic 1): 22 | High: 19 | Medium: 3

4. Story Point Estimation – Planning Poker

Story points were estimated independently by each team member using the Fibonacci scale (1, 2, 3, 5, 8, 13...). The individual estimates were compared, differences were discussed, and the team agreed on the final estimates.

Story ID	Story Title	Arfan	Kshama	Lakshanya	a Manjunatha	Agreed
SEL2-2 / 1.1	View Equipment Availability	3	3	5	3	3
SEL2-3 / 1.2	Check Slot Availability	5	5	3	5	5
SEL2-4 / 1.3	Verify Equipment Calibration	3	3	3	5	3
SEL2-5 / 1.4	Enforce Reservation Limits	3	5	3	3	3
SEL2-6 / 1.5	Lock Equipment Slot	5	5	5	3	5
SEL2-7 / 1.6	Generate Reservation Token	3	3	3	3	3
	Total					22 points

5. Sprint Planning & Simulation



Sprint 1 (Duration: 1 week)

Sprint Goal:
Enable students to identify available equipment and valid reservation slots while ensuring that selected equipment is calibrated and reservation limits are enforced.

Key	Story	Priority	Pts	Status
SEL2-2	1.1 View Equipment Availability	High	3	Done
SEL2-3	1.2 Check Slot Availability	High	5	Done
SEL2-4	1.3 Verify Equipment Calibration	High	3	Done
SEL2-5	1.4 Enforce Reservation Limits	High	3	Done

Sprint 1 Velocity:

14 pts committed | 14 pts completed | 0 pts rolled over

Sprint 2 — Reservation Confirmation

Sprint 2 Duration: 1 week

Sprint Goal:

Complete the reservation workflow by securely locking the selected equipment slot and generating a unique reservation token.

Key	Story	Priority	Pts	Status
SEL2-6	1.5 Lock Equipment Slot	Medium	5	Done
SEL2-7	1.6 Generate Reservation Token	Medium	3	Done

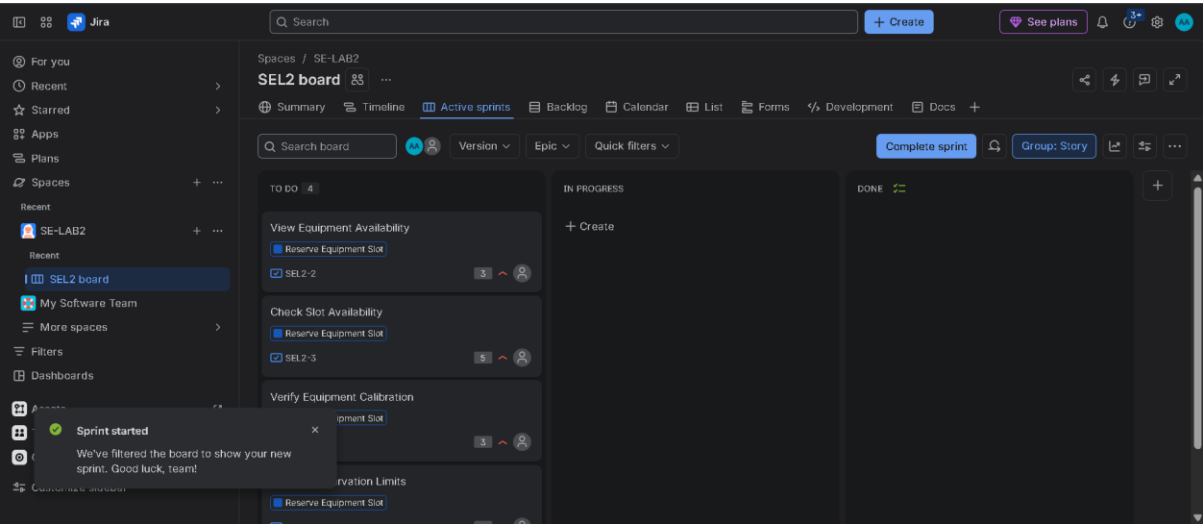
Sprint 2 Velocity:

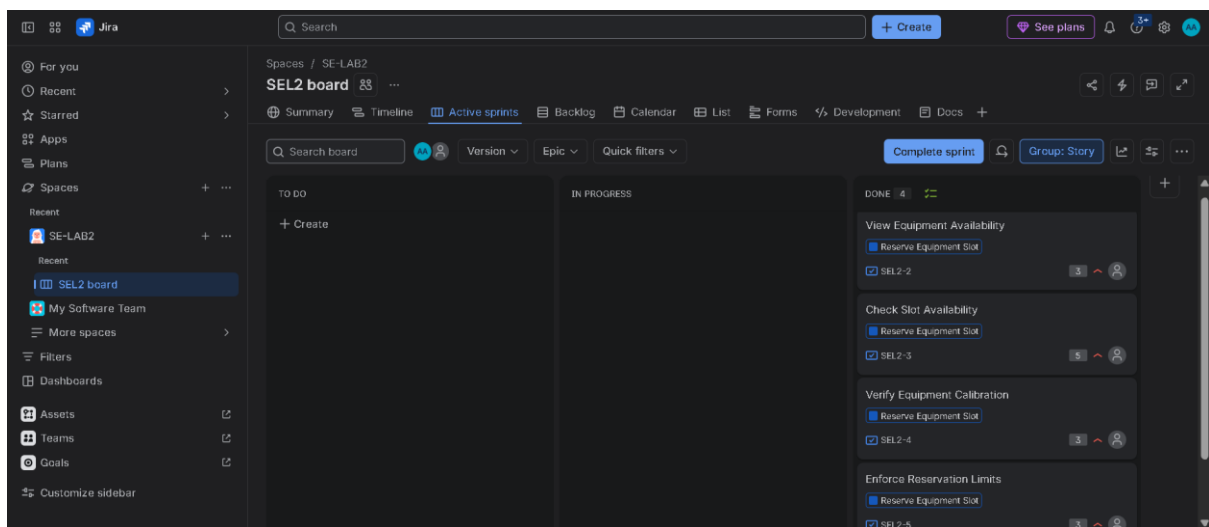
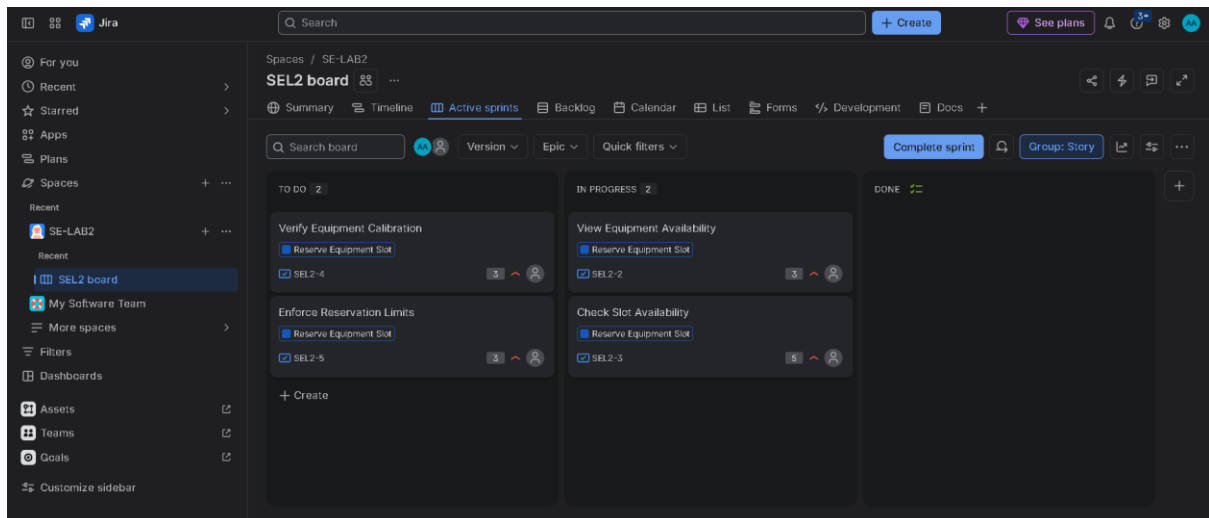
8 pts committed | 8 pts completed | 0 pts rolled over

Overall:

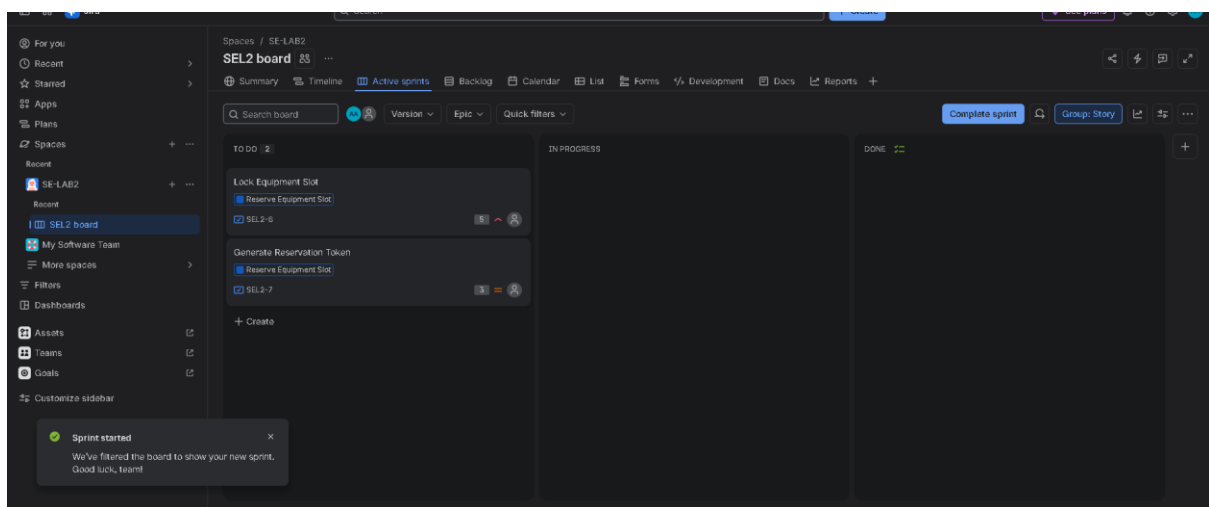
22 pts committed | 22 pts completed | 0 pts rolled over

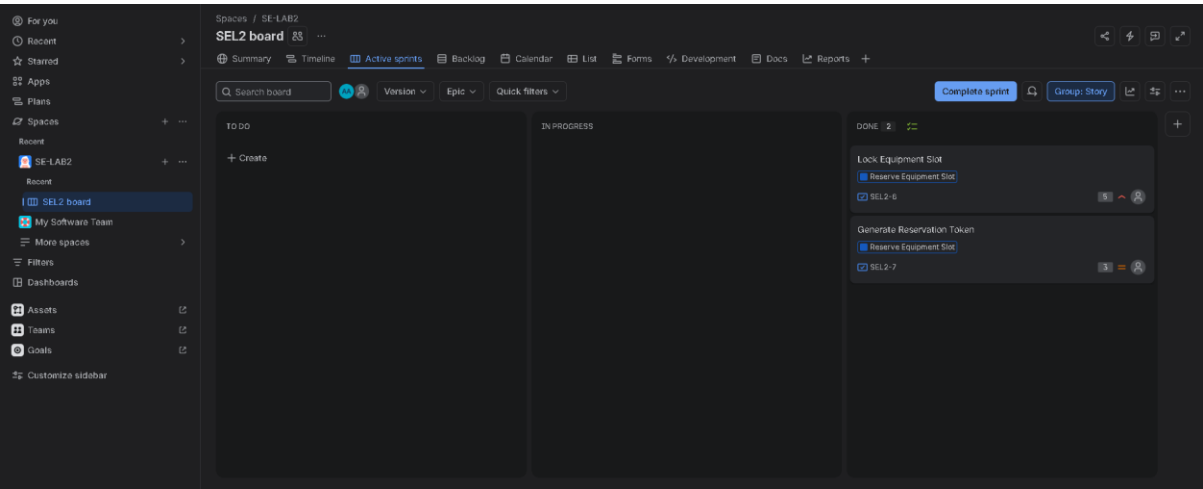
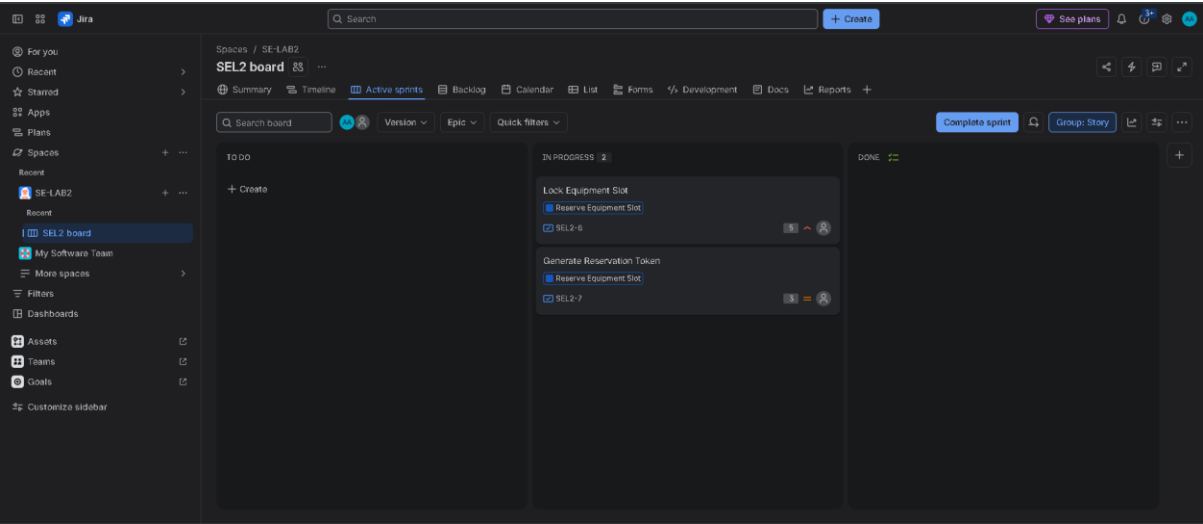
SPRINT 1 SCREENSHOTS





SPRINT 2 SCREENSHOTS





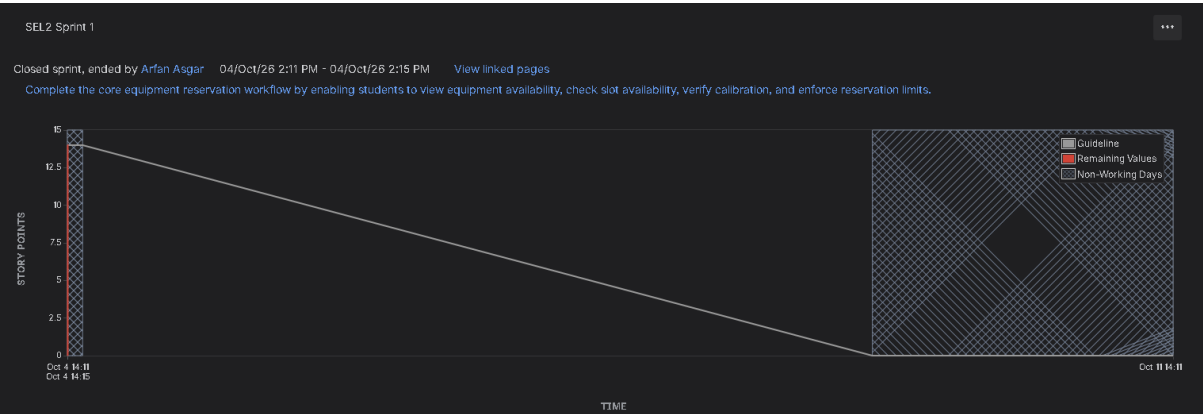
6. Burndown Chart Analysis

The Jira Burndown Chart plots story points remaining (red line) against the ideal guideline (grey line). Each sprint is shown twice: once with non-working days displayed (hatched) and once with them hidden, which makes the guideline easier to read. The event table under each chart lists the exact completion times.

Sprint 1 Burndown chart



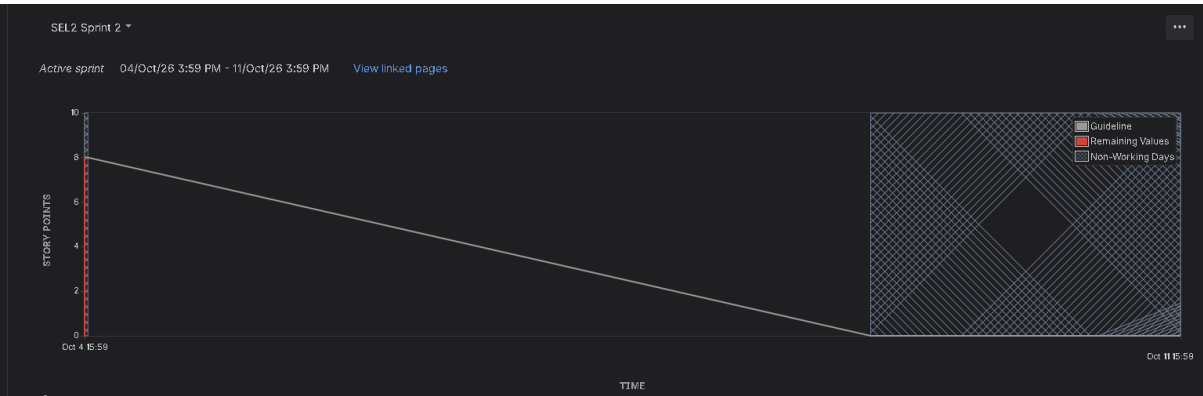
Sprint 1 Reports



Sprint 2 Burndown Chart



Sprint 2 Reports



Summary of the burndown data (from the Jira event tables)

Sprint	Start (points)	Completion events	End (points)
Sprint 1	14	SEL2-2 View Equipment Availability (-3) → 11; SEL2-3 Check Slot Availability (-5) → 6; SEL2-4 Verify Equipment Calibration (-3) → 3; SEL2-5 Enforce Reservation Limits (-3) → 0	0
Sprint 2	8	SEL2-6 Lock Equipment Slot (-5) → 3; SEL2-7 Generate Reservation Token (-3) → 0	0

7. Reflection Questions

1. Did your estimations reflect the actual effort?

Yes. The estimations were reasonably accurate because all the planned user stories were completed within their respective sprints without any work being carried over. Sprint 1 completed all **14 points**, while Sprint 2 completed all **8 points**. This indicates that the estimated story-point values were suitable for the simulated workload.

2. Was your backlog well-prioritized?

Yes. The backlog was prioritized according to the dependency of the reservation workflow. Basic availability, slot checking, calibration verification, and reservation-limit enforcement were placed in Sprint 1 because they form the foundation for a valid reservation. Slot locking and reservation-token generation were placed in Sprint 2 as the final steps of the reservation process.

3. How did your simulated sprint align with your plan?

The simulated sprints aligned well with the original plan. Sprint 1 was planned for **14 story points** and all 14 points were completed. Sprint 2 was planned for **8 story points** and all 8 points were completed. There were **0 points rolled over** from either sprint, showing that the planned workload was completed as intended.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed that the team was able to complete its committed work within the planned sprint duration. Sprint 1 reduced the workload from **14 to 0 points**, while Sprint 2 reduced it from **8 to 0 points**. This indicates that the simulated team had sufficient capacity to handle approximately **14–16 story points per one-week sprint**, with the actual workload in this simulation being completed successfully.